

Likelihood-Ranked Synthetic Selection in Self-Consuming Diffusion: Minority Extinction Ordering on Imbalanced Mixtures

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Abstract

Self-consuming training loops—retraining generative models on their own outputs—can induce *model collapse*, including loss of rare modes. Prior work notes that bias toward high-quality synthetic samples worsens the quality–diversity tradeoff, and that task-level verification can stabilize labeled synthetic data. We study the most common *unlabeled* selection score for diffusion models: a denoising ELBO / likelihood proxy. On imbalanced Gaussian mixtures with a tiny DDPM, under matched real-data mixing and matched synthetic budget, first-generation minority-mass retention obeys

$$\text{bottom-}k > \text{random-}k > \text{top-}k$$

(means 0.57/0.36/0.17 over eight seed–imbalance pairs; order holds on 5/5 seeds at imbalance $\rho=5$). Paired Wilcoxon tests vs. random- k give $p \approx 0.042$ for both top and bottom contrasts. Likelihood top- k also concentrates majority mass relative to random- k . A natural “false stability” hypothesis—that top- k improves sliced W_2 while minorities die—is falsified. The ordering transfers to 8D GMM and to Digits in PCA space with nearest class-mean occupancy (3/3 seeds each). We release configs, seeds, logs, and analysis code.

1 Introduction

As synthetic data proliferates, generative models are increasingly trained on mixtures of real and model-generated samples. Iterated self-consumption can cause *model collapse*: progressive loss of diversity and forgetting of rare modes [Shumailov et al., 2023, Alemohammad et al., 2023, Gerstgrasser et al., 2024]. Mitigations include accumulating real data [Gerstgrasser et al., 2024], injecting fresh real samples, and *filtering* synthetic data [Feng et al., 2024, Cai et al., 2025].

A practical question remains underspecified: when the only available score is the generator’s own likelihood (or denoising loss), how should one select synthetic samples? Feng et al. [2024] show that *task verifiers* for labeled synthetic answers can prevent collapse. Cai et al. [2025] filter “unrealistic” diffusion samples via latent probes and report aggregate FID/precision/recall. Alemohammad et al. [2023] observe that sampling bias toward high-quality generations drives a quality–diversity tradeoff. What is missing is a **matched-budget, mode-exact** causal contrast among likelihood ranks.

Contributions.

1. A controlled self-consuming DDPM protocol on imbalanced GMMs with exact mode masses.
2. Evidence for a **likelihood-rank ordering** of minority retention under matched k and matched real mix ratio.

3. An experimental **falsification** of W_2 -based false stability for likelihood top- k .
4. Transfer checks: 8D GMM and Digits-PCA with nearest class-mean occupancy.

2 Related work

Model collapse. Shumailov et al. [2023] document recursive training collapse and tail forgetting. Gerstgrasser et al. [2024] show accumulating real+synthetic data can avoid collapse. Alemohammad et al. [2023] (*MAD*) highlight quality-biased sampling.

Verification and filtering. Feng et al. [2024] argue verification of synthesized *labels* prevents collapse. Cai et al. [2025] (*LSF*) filter synthetic images using latent-space realism scores. Neither isolates unlabeled ELBO top- k vs. random- k with mode-grounded metrics on continuous mixtures.

Temperature / diversity knobs. Inference-time temperature methods [Xu et al., 2025, Li et al., 2026] reshape sampling diversity; we instead intervene on *training-set selection* inside self-consuming loops.

3 Method

3.1 Self-consuming loop

Let $f^{(g)}$ be a DDPM trained on $\mathcal{D}^{(g)}$. Sample synthetic set $\hat{\mathcal{D}}^{(g+1)}$. Build the next train set by mixing a fixed real buffer $\mathcal{D}_{\text{real}}$ (fraction α) with selected synthetics.

3.2 Likelihood proxy and policies

For each synthetic x , define score $s(x)$ as mean denoising MSE over a grid of times t (ELBO proxy; **lower** = higher likelihood). Policies at matched synthetic count: **top- k** (lowest s), **random- k** , **bottom- k** (highest s), plus **mix** / **replace** references.

3.3 Metrics

Nearest-mean mode assignment on known centers (GMM means or Digits empirical class means in PCA space). Report minority-mean mass, majority mass, and sliced W_2 to held-out real samples.

4 Experiments

Setup. $K=4$ modes, $\rho \in \{5, 10\}$, tiny MLP denoiser, $G=5$, seeds $\{0 \dots 4\}$ (seeds 3–4 replicate $\rho=5$), $\alpha=0.5$. Logs: `experiments/logs/w2_fs.jsonl` ($n=228$).

At $\rho=5$, generation 1, all five seeds satisfy bottom > rand > top. Wilcoxon signed-rank vs. random- k ($n=8$): top–rand $\Delta=-0.188$ ($p \approx 0.042$); bottom–rand $\Delta=+0.212$ ($p \approx 0.042$). Top- k does **not** improve sliced W_2 (mean $\Delta W_2 > 0$).

Real-mix sensitivity. An α -sweep (`w2_fs_alpha.jsonl`) shows the top- k penalty is clearest at $\alpha \leq 0.5$; at $\alpha=0.75$ it weakens vs. random while bottom- k remains highest on average.

Table 1: Minority-mass retention $m^{(1)}/m^{(0)}$ on the principal GMM matrix.

Policy	Mean	Std	n
bottom_k	0.572	0.289	8
rand_k	0.360	0.173	8
mix	0.328	0.100	8
top_k	0.172	0.078	8

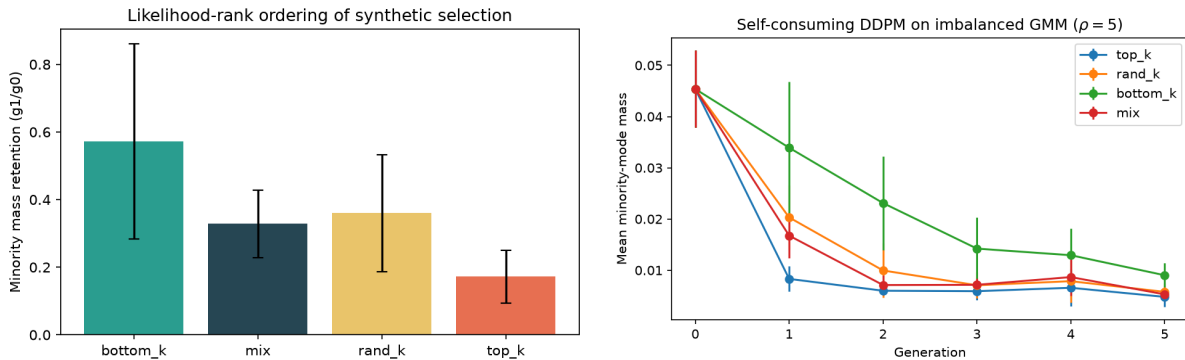


Figure 1: Left: retention by policy. Right: minority mass vs. generation at $\rho=5$.

Transfers. An 8D GMM transfer (`w2_fs_hd`) reproduces the ordering on 3/3 seeds (retention means 0.55/0.27/0.08). Digits in PCA-8 with nearest class-mean occupancy (`w2_fs_digits_pca`) holds on 3/3 seeds (retention means 1.49/1.11/0.68). An earlier Digits CVAE+classifier probe is inconclusive and unused for claims.

Selection-pool size. Under $k = \max(n_{\text{syn}}, \lfloor k_{\text{frac}} N \rfloor)$, $k_{\text{frac}} \in \{0.25, 0.5\}$ are operationally equivalent when n_{syn} binds; at $k_{\text{frac}}=0.75$ top- k remains worst.

5 Analysis

Likelihood top- k preferentially retains synthetic points the current model already explains well—typically majority-mode samples—reducing minority inflow into $\mathcal{D}^{(g+1)}$. Bottom- k does the opposite relative to random. This is consistent with MAD’s quality-bias narrative but makes the score-rank interventional and mode-exact. Aggregate W_2 is a poor monitor for this failure: it moves with top- k in the *wrong* direction for a “quality win.”

6 Limitations

Toy 2D/8D GMMs and Digits-PCA with tiny networks; not ImageNet-scale. Digits CVAE transfer was inconclusive; Digits-PCA uses embeddings, not raw pixels. Effect strongest at moderate imbalance and early generations; high real mix ($\alpha=0.75$) weakens the top- k vs. random gap. Novelty is **L2 empirical** relative to MAD/LSF/Feng; we do not claim a new SOTA training method on FID.

7 Conclusion

Unlabeled likelihood ranking of synthetic data in self-consuming diffusion induces a stable minority-retention order (bottom- k > random- k > top- k) under matched budgets on imbalanced mixtures, while failing to deliver W_2 improvements. Preferring “most likely” synthetics is not a free lunch for minority survival.

References

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A Reproducibility and code

Principal command (Windows PowerShell): `scripts/reproduce_w2.ps1`. All numerical tables are produced from logged JSONL via analysis scripts (no hand-edited metrics). Experiment IDs: `w2_fs`, `w2_fs_hd`, `w2_fs_digits_pca` (supported); `w2_fs_digits` (CVAE; inconclusive). Source code, configs, seeds, and logs accompany this manuscript in the project repository (same directory tree as the paper package).